Chloe Barker

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EDUCATION

Southern Methodist University | Dallas, TX Jan 2025 - Aug 2026
M.S. in Data Science | GPA: 3.95

The University of Texas at Austin | Austin, TX Aug 2020 - May 2024
B.S.A. in Biology | Minor: Health Communication | GPA: 3.77

TECHNICAL SKILLS

Programming & Analysis: Python (pandas, NumPy, scikit-learn), SQL, R (tidyverse, caret), PySpark, SAS

Machine Learning & Statistics: Linear & logistic regression, LASSO, random forest, gradient boosting (XGBoost, CatBoost), SVM, KNN, clustering, time series forecasting, NLP, feature engineering, cross-validation, hyperparameter tuning, model evaluation, hypothesis testing, A/B testing

Generative AI: LLM applications, retrieval-augmented generation (RAG), prompt engineering

Data Platforms & Engineering: Databricks, Snowflake, MySQL, ETL, data pipelines, data cleaning and transformation, relational database design, query optimization

Visualization: Tableau, Matplotlib, seaborn, Plotly, ggplot2, Folium, R Shiny

Tools & Productivity: Git/GitHub, Jupyter, HTML/CSS, Excel (PivotTables, data analysis), PowerPoint

EXPERIENCE

AAA Auto Club Enterprises | Coppell, TX Oct 2026 - Present
DevOps Support Analyst, Data Science Team

- Support the Data Science team's work with insurance data, building an understanding of business operations, key performance indicators, and existing data workflows.
- Assist with data pipeline validation, data quality monitoring, and documentation to support reliable reporting and analytics.
- Collaborate with team members on production support processes and the development of data-driven capabilities for insurance operations.

Goosehead Insurance | Westlake, TX May 2026 - Aug 2026
Data Science Intern

- Developed and evaluated machine-learning models on large-scale insurance data in Snowflake and Databricks to support lead prioritization, customer segmentation, and marketing targeting.
- Built end-to-end data preparation and modeling workflows in SQL, Python, and PySpark, engineering features and transforming raw policy and customer data into model-ready datasets.
- Analyzed classification performance and customer-level patterns, translating model results into actionable recommendations for data science and business stakeholders.

Outlier AI | Remote Aug 2024 - May 2026
AI Math Evaluator

- Evaluated AI-generated solutions across algebra, calculus, and statistics, identifying reasoning errors and assessing outputs for mathematical correctness, clarity, and completeness.

RUSH University | Chicago, IL Summers 2021 & 2023
Research Assistant, Dept. of Microbial Pathogens & Immunity

- Investigated alcohol, circadian, and HIV-medication effects on the gut microbiome and bone density using longitudinal clinical data in R and Excel.
- Prepared, visualized, and documented experimental data through reproducible workflows for multidisciplinary research teams.

SELECTED PROJECTS

M.S. Capstone: Groundwater Lithium & Dementia Prevalence | Python, GeoPandas, statsmodels

- Integrated 460K+ groundwater measurements with dementia prevalence and geospatial data across 254 Texas counties; modeled associations using OLS with HC3 robust errors, permutation testing, bootstrap confidence intervals, and sensitivity analyses.
- Developed an exposure-homogeneity framework to evaluate measurement uncertainty and mapped county-level geographic patterns; received the MSDS Best Capstone Presentation award.

Diabetes Transitional Care Decision Support | Python, scikit-learn, RAG, Gradio

- Built a Gradient Boosting risk-ranking system to prioritize high-risk diabetes discharges under limited transitional-care capacity; achieved 40.8% Precision@500 and 3.6× lift, capturing 204 readmissions among the top 500 patients.
- Integrated probability calibration, capacity and cost modeling, and source-grounded CMS/AHRQ RAG guidance; estimated +$113.7K modeled net value under baseline intervention assumptions.